

Predictive Customer Churn Model for a Telecom Company

Minor Project Report

Submitted By

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Domain

Artificial Intelligence

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Abstract

Customer churn is one of the major challenges faced by telecommunication companies, as losing existing customers directly affects revenue and increases customer acquisition costs. Predicting customers who are likely to leave allows companies to take preventive actions such as offering discounts, personalized plans, and improved customer support.

This project develops an end to end machine learning solution for predicting customer churn using the Telco Customer Churn dataset. The project includes data preprocessing, exploratory data analysis, feature engineering, model training, hyperparameter tuning, model evaluation, explainable AI using SHAP, and deployment through a Streamlit web application and REST API. Three machine learning models, Logistic Regression, Random Forest, and XGBoost, were trained and compared using multiple evaluation metrics to identify the most effective solution.

Introduction

Customer retention is an important factor for business growth in the telecommunications industry. Acquiring new customers is often much more expensive than retaining existing ones. Therefore, companies need intelligent systems that can identify customers who are at high risk of leaving.

Machine Learning provides powerful predictive models that can analyze customer behavior and identify patterns associated with churn. By using these predictions, businesses can implement targeted retention strategies and reduce customer loss.

This project demonstrates a complete machine learning workflow for customer churn prediction, starting from raw data and ending with a deployable prediction system.

Problem Statement

ConnectTel, a telecommunications company, experiences significant revenue loss due to customer churn. The company needs a predictive system that identifies customers who are likely to discontinue their services before they actually leave.

The primary objective of this project is to develop a machine learning model capable of accurately predicting customer churn while also providing explanations for the predictions using Explainable Artificial Intelligence techniques.

Objectives

The main objectives of this project are:

- Predict customer churn using supervised machine learning algorithms.
- Perform exploratory data analysis to identify important customer behavior patterns.

- Engineer new features that improve prediction accuracy.
- Compare multiple machine learning models.
- Optimize model performance using hyperparameter tuning.
- Interpret model predictions using SHAP.
- Deploy the trained model through a Streamlit web application and REST API.
- Provide actionable business recommendations for customer retention.

Dataset Description

The project uses the Telco Customer Churn Dataset, which contains customer information collected from a telecommunications company.

The dataset includes customer demographic information, subscribed services, account details, billing information, and customer churn status.

Attribute	Value
Dataset	Telco Customer Churn
Total Records	7043
Features	21
Target Variable	Churn
Source	Kaggle

Important Features

- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Phone Service
- Internet Service
- Online Security
- Streaming TV
- Streaming Movies

- Contract Type
- Payment Method
- Monthly Charges
- Total Charges
- Churn (Target Variable)

The target variable is **Churn**, where:

- Yes = Customer Left
- No = Customer Stayed

6. Data Preprocessing

```
*** Raw Dataset Shape: (7043, 21)

Initial Column Datatypes:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                7043 non-null   object
2   SeniorCitizen         7043 non-null   int64
3   Partner               7043 non-null   object
4   Dependents            7043 non-null   object
5   tenure               7043 non-null   int64
6   PhoneService          7043 non-null   object
7   MultipleLines         7043 non-null   object
8   InternetService       7043 non-null   object
9   OnlineSecurity        7043 non-null   object
10  OnlineBackup          7043 non-null   object
11  DeviceProtection      7043 non-null   object
12  TechSupport           7043 non-null   object
13  StreamingTV           7043 non-null   object
14  StreamingMovies       7043 non-null   object
15  Contract              7043 non-null   object
16  PaperlessBilling      7043 non-null   object
17  PaymentMethod         7043 non-null   object
18  MonthlyCharges        7043 non-null   float64
19  TotalCharges          7043 non-null   object
20  Churn                 7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

Converted TotalCharges Datatype: float64
Total Missing Values: 0
Total Duplicate Rows: 22
```

Data preprocessing is an essential step in building an accurate machine learning model. The Telco Customer Churn dataset was first imported into Python using the Pandas library. The dataset was examined for missing values, duplicate records, and inconsistent data types.

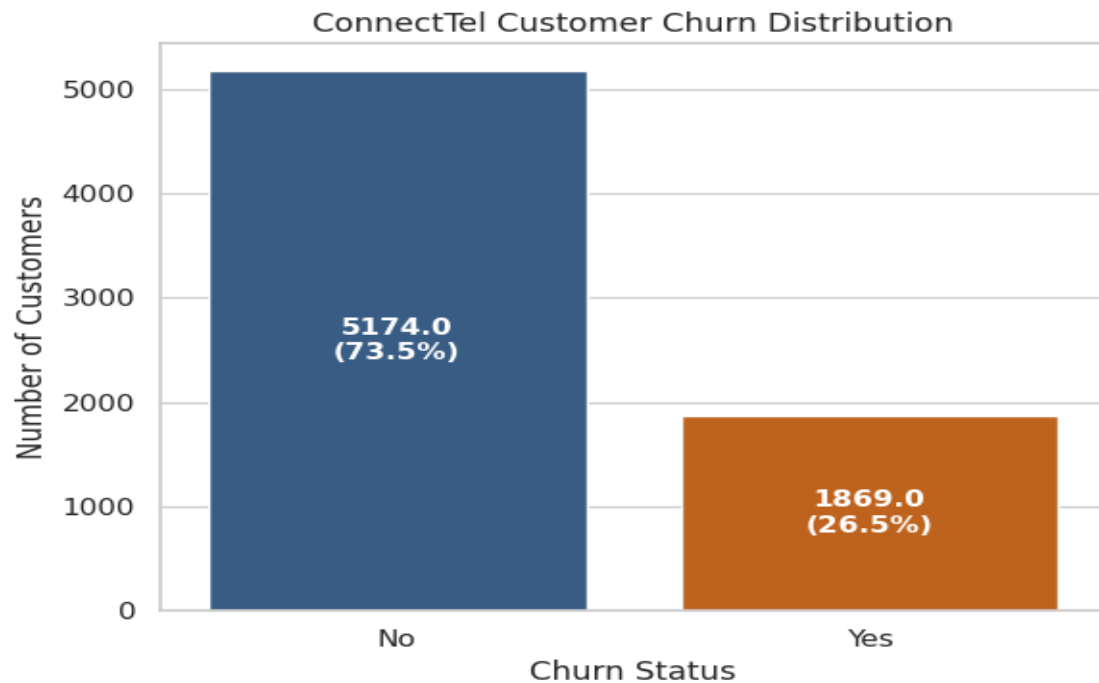
The **TotalCharges** column contained a few missing values due to blank entries. These missing values were handled appropriately to ensure data quality. Categorical features such as **Gender**, **Partner**, **Dependents**, **Contract**, **InternetService**, and **PaymentMethod** were converted into numerical values using **One Hot Encoding**. This allowed machine learning algorithms to process categorical information efficiently.

Numerical features including **Tenure**, **MonthlyCharges**, and **TotalCharges** were standardized using **StandardScaler** to improve model performance. Finally, the processed dataset was divided into training and testing sets using an 80:20 ratio.

7. Exploratory Data Analysis (EDA)

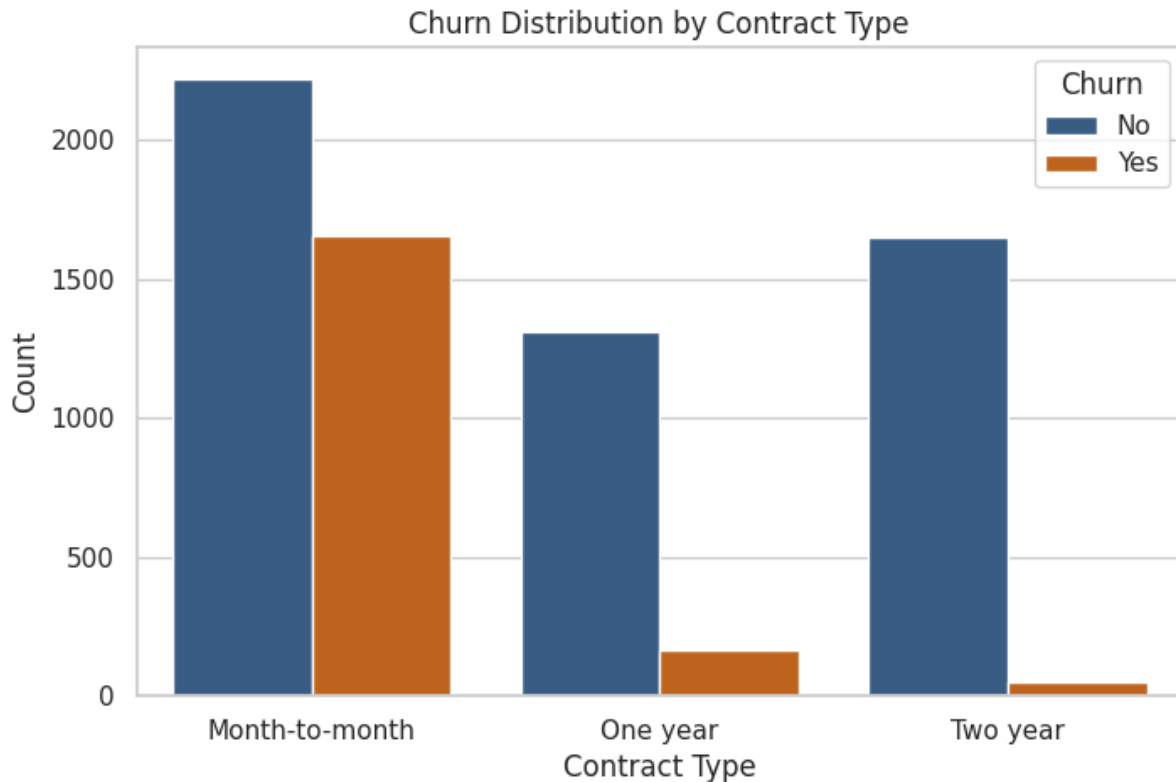
Exploratory Data Analysis was performed to understand customer behavior and identify factors influencing churn. Several visualizations were created to analyze relationships between customer attributes and the target variable.

Figure 1: Customer Churn Distribution



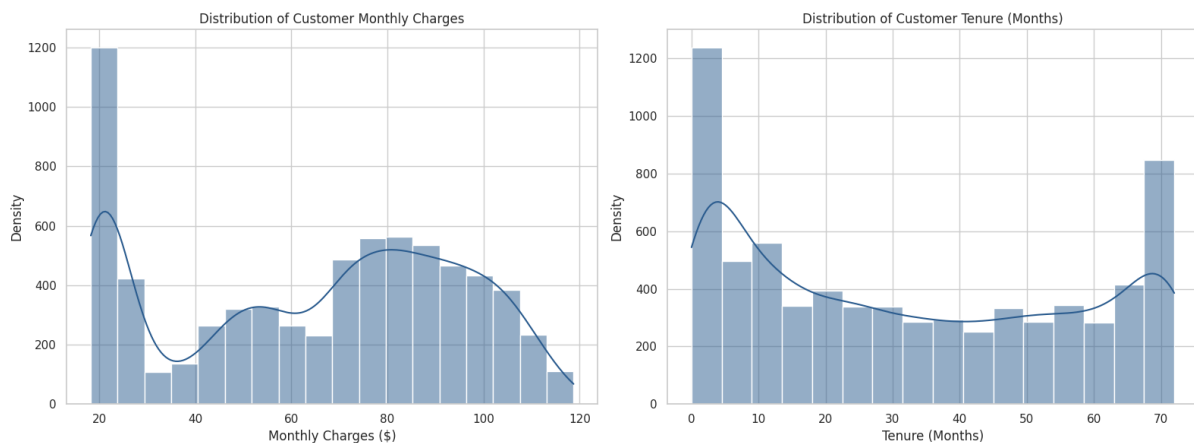
The graph shows that the number of customers who stayed is higher than those who churned, indicating that the dataset is moderately imbalanced.

Figure 2: Contract Type vs Customer Churn



Customers with **Month-to-Month** contracts showed the highest churn rate, while customers with **One-Year** and **Two-Year** contracts were significantly more likely to remain with the company.

Figure 3: Monthly Charges Distribution



Customers with higher monthly charges were generally more likely to churn, suggesting that pricing may influence customer decisions.

Figure 4: Tenure Distribution

Customers with shorter tenure were more likely to leave, whereas long-term customers demonstrated greater loyalty.

The EDA results provided valuable insights that guided feature engineering and model development.

8. Feature Engineering

Feature engineering was performed to improve the predictive capability of the models. New variables were created using existing customer information.

Engineered Feature	Description
ServiceCount	Counts the total number of subscribed services for each customer.
TotalChargesPerTenure	Calculates the average amount spent by a customer during their subscription period.

9. Machine Learning Models

Three different machine learning algorithms were implemented and compared.

Logistic Regression

Logistic Regression served as the baseline classification model. It is simple, computationally efficient, and provides easily interpretable results for binary classification problems.

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy while reducing overfitting. It effectively captures complex relationships within the data.

XGBoost

XGBoost is an advanced gradient boosting algorithm widely recognized for its excellent performance on structured datasets. Hyperparameter tuning was applied to maximize prediction performance.

Cross-validation and GridSearchCV were used to identify the optimal parameters for each model.

10. Model Evaluation

The performance of all models was evaluated using Accuracy, Precision, Recall, F1 Score, and ROC-AUC.

Table 1: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Logistic Regression	73.81%	50.42%	80.48%	62.00%	84.61%
Tuned Random Forest	75.30%	52.35%	77.54%	62.50%	83.99%
Tuned XGBoost	74.38%	51.11%	79.95%	62.36%	84.43%

Analysis

Among the three evaluated machine learning models, **Logistic Regression achieved the highest ROC-AUC score of 84.61%**, indicating the best ability to distinguish between customers who are likely to churn and those who are not. Although the **Tuned Random Forest** obtained the highest overall accuracy of **75.30%**, its ROC-AUC was slightly lower than Logistic Regression. The **Tuned XGBoost** model also delivered competitive performance with a ROC-AUC of **84.43%**.

Since customer churn prediction involves an imbalanced dataset, **ROC-AUC, Precision, Recall, and F1 Score** are more reliable evaluation metrics than accuracy alone. Based on these metrics, **Logistic Regression was selected as the final model**, as it provides the best balance between identifying churning customers and minimizing misclassification.

Confusion Matrix

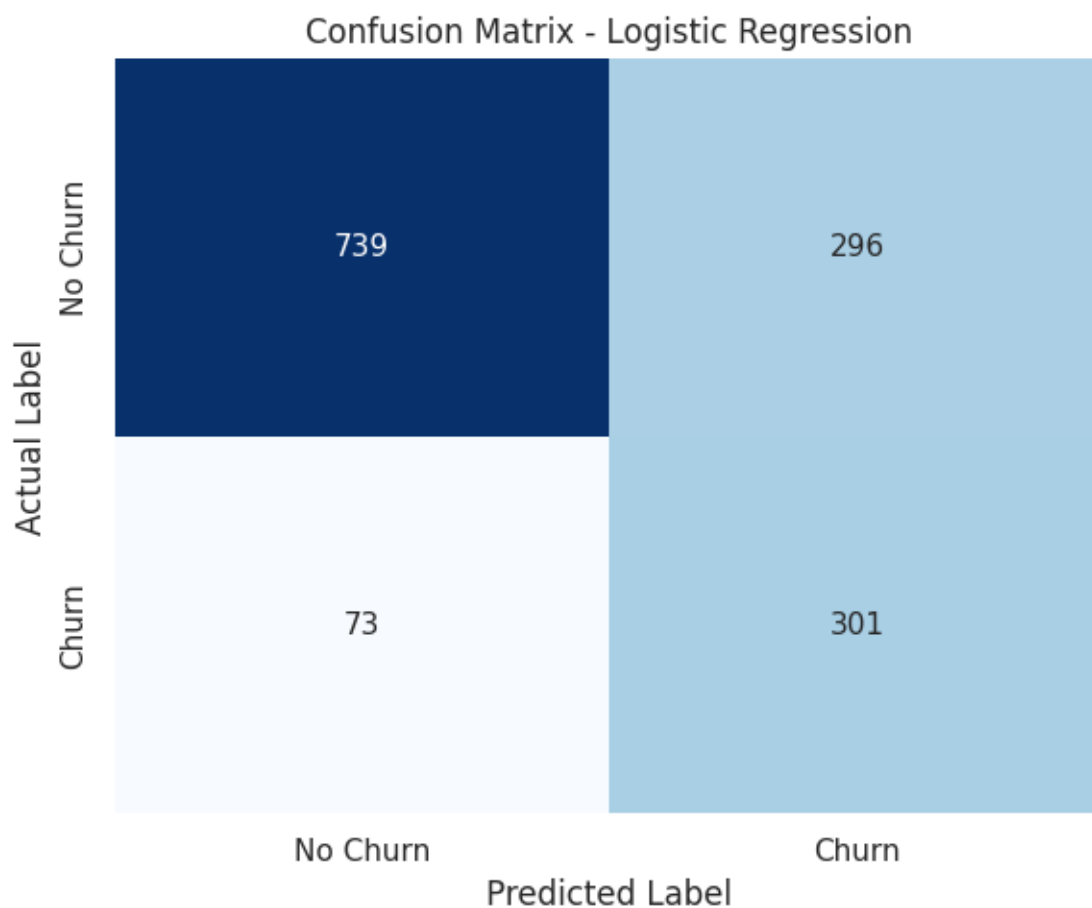
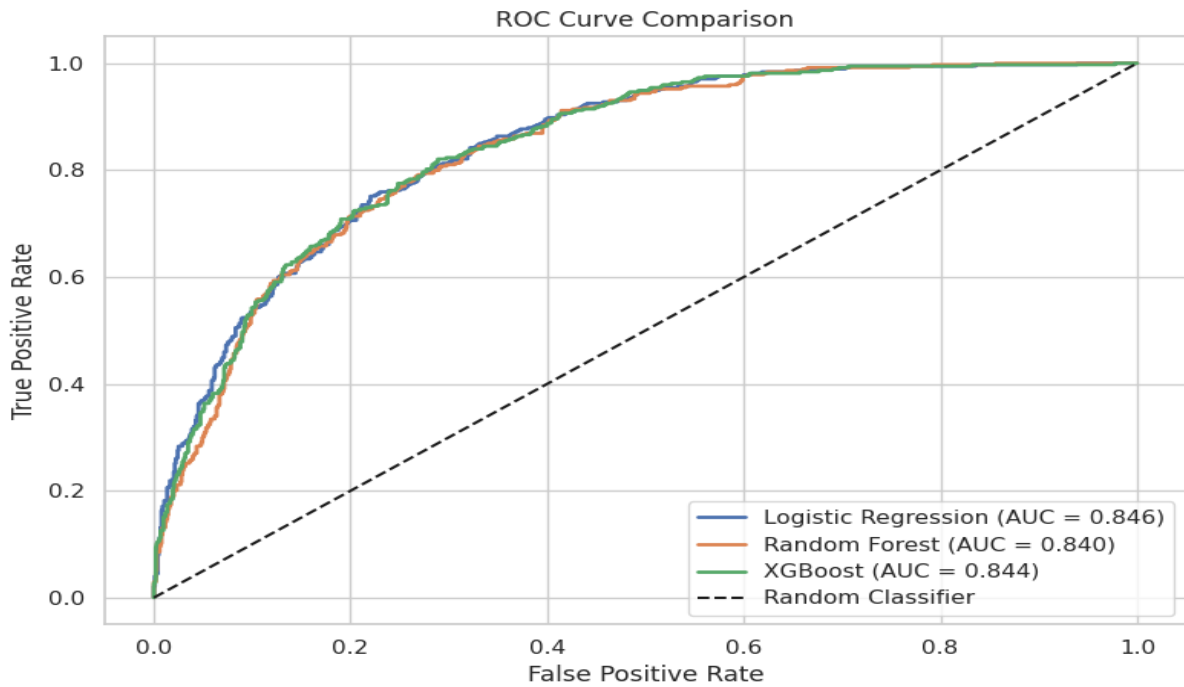


Figure 5 shows the confusion matrix for the Logistic Regression model. The model correctly classified 739 non-churn customers and 301 churn customers, while 296 non-churn customers were incorrectly predicted as churn and 73 churn customers were incorrectly predicted as non-churn. These results demonstrate that the model achieves high recall for churn prediction, making it suitable for identifying customers who are at risk of leaving.

ROC Curve

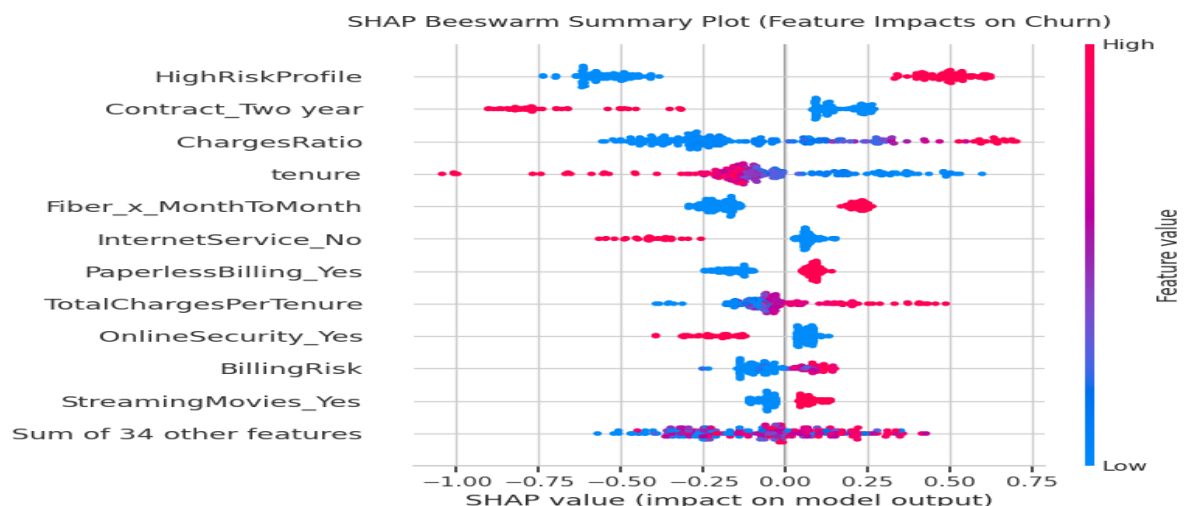


Logistic Regression achieved the highest ROC-AUC score of 84.61%, indicating the best overall balance between correctly identifying customers who are likely to churn and minimizing false predictions.

11. Model Interpretability Using SHAP

Although advanced machine learning models provide high prediction accuracy, understanding why a prediction is made is equally important. SHAP (SHapley Additive exPlanations) was used to interpret the model.

SHAP Summary Plot



The SHAP analysis revealed that the most influential features affecting customer churn include:

- Contract Type

- Monthly Charges
- Tenure
- Internet Service
- Total Charges

Customers with month-to-month contracts, higher monthly charges, and shorter tenure were identified as having a higher probability of churn.

12. Business Recommendations

Based on the analysis and model predictions, the following recommendations are suggested for ConnectTel:

- Encourage customers to switch from month-to-month contracts to yearly subscription plans.
- Offer personalized discounts to customers with high monthly charges.
- Improve service quality and customer support for Fiber Optic users.
- Monitor newly joined customers and provide early engagement offers.
- Use the developed churn prediction model regularly to identify high-risk customers and initiate retention campaigns before they leave.

Implementing these recommendations can reduce customer churn, improve customer satisfaction, and increase long-term revenue.

13. Conclusion

This project successfully developed an end-to-end machine learning solution for predicting customer churn in a telecommunications company. The workflow included data preprocessing, exploratory data analysis, feature engineering, model development, hyperparameter tuning, model evaluation, and explainable AI using SHAP.

Among the evaluated machine learning algorithms, Logistic Regression achieved the highest ROC-AUC score of 84.61% and was selected as the final model. The developed system successfully predicts customer churn while providing interpretable predictions using SHAP analysis. The deployment through Streamlit and REST API demonstrates the practical applicability of the solution in supporting customer retention strategies.

14. Future Scope

The project can be further enhanced in several ways:

- Integrate real-time customer data for continuous prediction.
- Develop deep learning models for improved accuracy.
- Incorporate customer support interactions and complaint history.

- Deploy the solution on cloud platforms for large-scale production use.
- Build an automated customer retention recommendation system using predictive analytics.

These improvements can make the system more scalable, intelligent, and effective in real-world business environments.